18.1: Bacteria often respond to environmental change by regulating transcription

Bacterial cells that can conserve resources and energy have a selective advantage over cells that are unable to do so. Thus, natural selection has favored bacteria that express only the genes whose products are needed by the cell.

Consider, for instance, an individual *Escherichia coli* (*E. coli*) cell living in a human colon, dependent for its nutrients on the whimsical eating habits of its host. If the environment is lacking in the amino acid tryptophan, which the bacterium needs to survive, the cell responds by activating a metabolic pathway that makes tryptophan from another compound. If the human host later eats a tryptophan-rich meal, the bacterial cell stops producing tryptophan, thus avoiding wasting resources to produce a substance that is readily available from the surrounding solution.

A metabolic pathway can be controlled on two levels, as shown for the synthesis of tryptophan in **Figure 18.2**. First, cells can adjust the activity of enzymes already present. This is a fairly rapid physiological response, which relies on the sensitivity of many enzymes to chemical cues that increase or decrease their catalytic activity (see Concept 8.5). The activity of the first enzyme in the pathway is inhibited by the pathway's end product—tryptophan, in this case (Figure 18.2a). Thus, if tryptophan accumulates in a cell, it shuts down the synthesis of more tryptophan by inhibiting enzyme activity. Such *feedback inhibition*, typical of anabolic (biosynthetic) pathways, allows a cell to adapt to short-term fluctuations in the supply of a substance it needs (see Figure 8.21).

Figure 18.2

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Regulation of a metabolic pathway.
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Regulation of a metabolic pathway.

In the pathway for tryptophan synthesis, an abundance of tryptophan can both ***(a)*** *inhibit the activity of the first enzyme in the pathway (feedback inhibition), a rapid response, and* ***(b)*** *repress expression of the genes encoding all subunits of the enzymes in the pathway, a longer-term response. The minus sign in a red circle stands for inhibition.*

Part a. Regulation of enzyme activity. Flow chart depicts the metabolic pathway for the synthesis of tryptophan. A precursor goes to the first product with enzyme 1. 1st product goes to the second product with enzyme 2. 2nd product goes to third product with enzyme 3. The end result of the pathway is tryptophan.

The presence of tryptophan can inhibit the first enzyme in the pathway in feedback inhibition. Part b. Regulation of enzyme production. Genes responsible for the production of enzymes in the metabolic pathway that synthesizes tryptophan (18.2a) are detailed:

t r p E and t r p D encode Enzyme 1.

t r p C encodes Enzyme 2.

t r p B and t r p A encode Enzyme 3.

Second, cells can adjust the production level of certain enzymes via a genetic mechanism; that is, they can regulate the expression of the genes encoding the enzymes. If, in our example, the environment provides all the tryptophan the cell needs, the cell stops making the enzymes that catalyze the synthesis of tryptophan (Figure 18.2b). In this case, the control of enzyme production occurs at the level of transcription, the synthesis of messenger RNA from the genes that code for these enzymes.

Regulation of the tryptophan synthesis pathway is just one example of how bacteria tune their metabolism to changing environments. Many genes of the bacterial genome are switched on or off by changes in the metabolic status of the cell; some genes are regulated singly and others as groups of related genes. One basic mechanism for this type of regulation of groups of genes in bacteria, described as the *operon model*, was discovered in 1961 by François Jacob and Jacques Monod at the Pasteur Institute in Paris. Let's see what an operon is and how it works.

Operons: The Basic Concept

E. coli synthesizes the amino acid tryptophan from a precursor molecule in the three-step pathway shown in Figure 18.2. Each reaction in the pathway is catalyzed by a specific enzyme, and the five genes that code for the subunits of these enzymes are clustered together on the bacterial chromosome. A single promoter serves all five genes, which together constitute a transcription unit. (Recall that a promoter is a site where RNA polymerase can bind to DNA and begin transcription; see Figure 17.8.) Thus, transcription gives rise to one long mRNA molecule that codes for the five polypeptides making up the enzymes in the tryptophan pathway **(Figure 18.3a)**. The cell can translate this one mRNA into five separate polypeptides because the mRNA is punctuated with start and stop codons that signal where the coding sequence for each polypeptide begins and ends.

Figure 18.3

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The *trp* operon in *E. coli*: regulated synthesis of
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repressible enzymes.
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The t r p operon in E. coli.

Part a. Tryptophan absent, repressor inactive, operon on. R N A polymerase attaches to the D N A at the operon's promoter and transcribes the operon's genes. In the D N A, t r p R (regulatory gene) with its own promoter is transcribed into m R N A and translated into a protein that is the inactive repressor. Binding sites on the repressor protein are empty. The Trp operon is a region of the D N A that includes (from left to right): the promoter, which includes the R N A polymerase and the operator, followed by operon genes t r p E, t r p D, t r p C and t r p A. As the R N A polymerase works from left to right, a long, continuous strand of m R N A is transcribed from the operon genes with start and stop codons for each gene. Following transcription, operon genes are translated into polypeptide subunits that make up enzymes from tryptophan synthesis (see previous figure). Those polypeptide subunits are: E (from t r p E); D (from t r p D); C (from t r p C); B (from t r p B) and A (from t r p A).

Part b. Tryptophan present, repressor active, operon off. As tryptophan accumulates, it inhibits its own production by activating the repressor protein, which binds to the operator, blocking transcription. The regulatory gene (t r p R) is transcribed in m R N A and translated into the repressor protein. Tryptophan, a corepressor, binds to the repressor protein, which becomes an active repressor. This active repressor binds to the operator in the promoter region of the t r p operon, which blocks transcription (R N A from being made). R N A polymerase can not bind.

Tryptophan is an amino acid produced by an anabolic pathway catalyzed by three enzymes (see Figure 18.2). **(a)** The five genes encoding the polypeptide subunits of the enzymes in this pathway are grouped, along with a promoter, into the *trp* operon. The trp operator (the repressor binding site) is located within the *trp* promoter (the RNA polymerase binding site). **(b)** Accumulation of tryptophan, the end product of the pathway, represses transcription of the *trp* operon, thus blocking synthesis of all the enzymes in the pathway and shutting down tryptophan production.

VISUAL SKILLS *Describe what happens to the* trp *operon as the cell uses up its store of tryptophan.*

*For suggested answer, see*Appendix A*.*

A key advantage of grouping genes of related function into one transcription unit is that a single "on-off switch" can control the whole cluster of functionally related genes; in other words, these genes are *coordinately controlled*. When an *E. coli* cell must make tryptophan for itself because its surrounding environment lacks this amino acid, all the enzymes for the metabolic pathway are synthesized at the same time. The on-off switch is a segment of DNA called an **operator**. Both its location and name suit its function: Positioned within the promoter or, in some cases, between the promoter and the enzyme-coding

genes, the operator controls the access of RNA polymerase to the genes. Together, the operator, the promoter, and the genes they control—the entire stretch of DNA required for enzyme production for the tryptophan pathway—constitute an **operon**. The *trp* operon (*trp* for tryptophan) is one of many operons in the *E. coli* genome (see Figure 18.3a).

If the operator is the operon's switch for controlling transcription, how does this switch work? By itself, the *trp* operon is turned on; that is, RNA polymerase can bind to the promoter and transcribe the genes of the operon. The *trp* operon can be switched off by a protein that is called the *trp* repressor. A **repressor** binds to the operator, preventing RNA polymerase from transcribing the genes, often by preventing RNA polymerase from binding **(Figure 18.3b)**. A repressor protein is specific for the operator of a particular operon. For example, the *trp* repressor, which switches off the *trp* operon by binding to the *trp* operator, has no effect on other operons in the *E. coli* genome.

A repressor protein is encoded by a **regulatory gene**—in this case, a gene called *trpR; trpR* is located some distance from the *trp* operon and has its own promoter. Regulatory genes are among the bacterial genes that are expressed continuously, although at a low rate, and a few *trp* repressor molecules are always present in *E. coli* cells. Why, then, is the *trp* operon not switched off permanently? First, the binding of repressors to operators is reversible. An operator alternates between two states: one with the repressor bound and one without the repressor bound. The relative duration of the repressor-bound state is higher when more active repressor molecules are present. Second, the *trp* repressor, like most regulatory proteins, is an allosteric protein, with two alternative shapes: active and inactive (see Figure 8.20). The *trp* repressor is synthesized in the inactive form, which has little affinity for the *trp* operator. Only when a tryptophan molecule binds to the *trp* repressor at an allosteric site does the repressor protein change to the active form that can attach to the operator, turning the operon off.

Tryptophan functions in this system as a **corepressor**, a small molecule that cooperates with a repressor protein to switch an operon off. As tryptophan accumulates, more tryptophan molecules associate with *trp* repressor molecules, which can then bind to the *trp* operator and shut down production of the tryptophan pathway enzymes. If the cell's tryptophan level drops, many fewer *trp* repressor proteins would have tryptophan bound, rendering them inactive; they would dissociate from the operator, allowing transcription of the operon's genes to resume. The *trp* operon is one example of how gene expression can respond to changes in the cell's internal and external environment.

Repressible and Inducible Operons: Two Types of Negative Gene Regulation

The *trp* operon is said to be a *repressible operon* because its transcription is usually on but can be inhibited (repressed) when a specific small molecule (in this case, tryptophan) binds allosterically to a regulatory protein. In contrast, an *inducible operon* is usually off but can be stimulated (induced) to be on when a specific small molecule interacts with a different regulatory protein. The classic example of an inducible operon is the *lac* operon (*lac* stands for "lactose").

The disaccharide lactose (milk sugar) is available to *E. coli* when the bacterium is in contact with any dairy product. Lactose metabolism by *E. coli* begins with hydrolysis of the disaccharide into its component monosaccharides (glucose and galactose), a reaction catalyzed by the enzyme β-galactosidase. Only a few molecules of this enzyme are present in an *E. coli* cell growing in the absence of lactose. If lactose is added to the bacterium's environment, however, the number of β-galactosidase molecules in the cell increases 1,000-fold within about 15 minutes. How can a cell ramp up enzyme production this quickly?

The gene for β-galactosidase (*lacZ*) is part of the *lac* operon, which includes two other genes coding for enzymes that function in the use of lactose **(Figure 18.4)**. The entire transcription unit is under the command of one main operator and promoter. The regulatory gene, *lacI*, located outside the *lac* operon, codes for an allosteric repressor protein that can switch off the *lac* operon by binding to the *lac* operator. So far, this sounds just like regulation of the *trp* operon, but there is one important difference. Recall that the *trp* repressor protein is inactive by itself and requires tryptophan as a corepressor in order to bind to the operator. The *lac* repressor, in contrast, is active by itself, binding to the operator and switching the *lac* operon off. In this case, a specific small molecule, called an **inducer**, *inactivates* the repressor.

Figure 18.4

The *lac* operon in *E. coli*: regulated synthesis of inducible enzymes.

The lac operon in E. coli.

Parts are as follows.

Part a. Lactose absent, repressor active, operon off. The lac repressor is innately active, and in the absence of lactose it switches off the operon by binding to the operator. In the D N A, the E. coli regulatory gene and the lac operon are diagramed. The lac operon consists of a promoter region that includes a teal region whose function will be revealed in Figure 18.5, followed by the operator, and then the genes of the operon. Only l a c Z is shown. Upstream (left) of the promoter is the regulatory gene with its own promoter region. The regulatory gene is transcribed into m R N A that is translated into a protein that is an active repressor. The repressor binds to the operator in the lac operon, blocking the R N A polymerase from binding and transcription does not occur. No R N A is made.

Part b. Lactose present, repressor inactive, operon on. Allolactose, an isomer of lactose, depresses the operon by inactivating the repressor. In this way, the enzymes for lactose utilization are induced. Diagram of the lac

operon, from left to right: Promoter (teal region, green promoter region); operator; l a c Z, l a c Y, l a c A. Upstream (left) of the promoter is the regulatory gene l a c I. This gene is transcribed into m R N A and translated into a repressor protein. Allolactose, the inducer, is present, and it binds to the protein, making it an inactive repressor. R N A polymerase can bind to the promoter region (green) and transcription of the lac operon occurs left to right. A continuous strand of m R N A is transcribed with start and stop codons for each of the operon genes. The results of translation for each gene transcribed from the operon are: l a c Z: Beta-galactosidase; l a c Y: Permease; l a c A: Transacetylase.

E. coli uses three enzymes to take up and metabolize lactose, the genes for which are clustered in the lac operon. The first gene, *lacZ*, codes for β-galactosidase, which hydrolyzes lactose to glucose and galactose. The second, *lacY*, codes for a permease, the membrane protein that transports lactose into the cell. The third, *lacA*, codes for transacetylase, an enzyme that detoxifies other molecules entering the cell via the permease. Unusually, the gene for the lac repressor, *lacI*, is adjacent to the *lac* operon. The function of the teal region within the promoter will be revealed in Figure 18.5.

Animation

The *lac* Operon in *E. coli*

Watch an explanation of this figure in the Figure 18.4 Walkthrough: The The *lac* operon in *E. coli*: regulated synthesis of inducible enzymes.

image

Play

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Play

For the *lac* operon, the inducer is allolactose, an isomer of lactose formed in small amounts from lactose that enters the cell. In the absence of lactose (and therefore allolactose), the *lac* repressor is in its active shape and binds to the operator; thus, the genes of the *lac* operon are silenced (Figure 18.4a). If lactose is added to the cell's surroundings, allolactose binds to the *lac* repressor and alters its shape so the repressor can no longer bind to the operator. Without the *lac* repressor bound, the *lac* operon is transcribed into mRNA, and the enzymes for using lactose are made (Figure 18.4b).

In the context of gene regulation, the enzymes of the lactose pathway are referred to as *inducible enzymes* because their synthesis is induced by a chemical signal (allolactose, in this case). Analogously, the enzymes for tryptophan synthesis are said to be repressible. *Repressible enzymes* generally function in anabolic pathways, which synthesize essential end products from raw materials (precursors). By suspending production of an end product when it is already present in sufficient quantity, the cell can allocate its organic

precursors and energy for other uses. In contrast, inducible enzymes usually function in catabolic pathways, which break down a nutrient to simpler molecules. By producing the appropriate enzymes only when the nutrient is available, the cell avoids wasting energy and precursors making proteins that are not needed.

Regulation of both the *trp* and *lac* operons involves the *negative* control of genes because the operons are switched off by the active form of their respective repressor protein. It may be easier to see this for the *trp* operon, but it is also true for the *lac* operon. In the case of the *lac* operon, allolactose induces enzyme synthesis not by directly activating the *lac* operon, but by freeing it from the negative effect of the repressor (see Figure 18.4b). Gene regulation is said to be *positive* only when a regulatory protein interacts directly with the genome to increase transcription.

Positive Gene Regulation

When glucose and lactose are both present in its environment, *E. coli* preferentially uses glucose. The enzymes for glucose breakdown in glycolysis (see Figure 9.8) are continually present. Only when lactose is present *and* glucose is in short supply does *E. coli* use lactose as an energy source, and only then does it synthesize appreciable quantities of the enzymes for lactose breakdown.

How does the *E. coli* cell sense the glucose concentration and relay this information to the *lac* operon? Again, the mechanism depends on the interaction of an allosteric regulatory protein with a small organic molecule, ****__cyclic AMP (cAMP)__ in this case, which accumulates when glucose is scarce (see Figure 11.11 for the structure of cAMP). The regulatory protein, called *cAMP receptor protein* (*CRP*), is an **activator**, a protein that binds to DNA and stimulates transcription of a gene. (CRP is also called *catabolite activator protein*, or CAP.) When cAMP binds to this regulatory protein, CRP assumes its active shape and can attach to a specific site at the upstream end of the *lac* promoter **(Figure 18.5a)**. This attachment increases the affinity of RNA polymerase for the *lac* promoter, which is actually rather low even when no *lac* repressor is bound to the operator. By facilitating the binding of RNA polymerase to the promoter and thereby increasing the rate of transcription of the *lac* operon, the attachment of CRP to the promoter directly stimulates gene expression. Therefore, this mechanism qualifies as positive regulation.

Figure 18.5

Positive control of the lac operon by cAMP receptor protein (CRP).

Positive control of the lac operon by c A M P receptor protein (C R P).

RNA polymerase has high affinity for the lac promoter only when CRP is bound to a DNA site at the upstream end of the promoter. CRP, in turn, attaches to its DNA site only when associated with cyclic AMP (cAMP), whose concentration in the cell rises when the glucose concentration falls. Thus, when glucose is

present, even if lactose is also available, the cell preferentially catabolizes glucose and makes very low levels of the enzymes for using lactose.

Parts are as follows.

Part a. Lactose present, glucose scarce (c A M P level high): abundant lac m R N A synthesized. If glucose is scarce, the high level of c A M P activates CAP, and the lac operon produces large amounts of m R N A coding for enzymes in the lactose pathway. Diagram of lac operon and regulatory gene. Lac I is upstream of lac operon. Lac operon is (left to right): promoter region (both teal and green promoter regions included); operator; genes of the operon. Diagram also shows presence of inactive lac repressor (with allolactose bound) and CAP, which is activated by the binding of c A M P. Active CAP can then bind to the teal region at the upstream end of the lac promoter. R N A polymerase binds to the promoter and transcribes the genes in the lac operon.

Part b. Lactose present, glucose present (c A M P level low): little lac m R N A synthesized. When glucose is present, c A M P is scarce, and CAP is unable to stimulate transcription at a significant rate, even though no repressor is bound. Diagram shows lac operon and regulatory gene. Inactive lac repressor is present with allolactose bound. Inactive CAP does not have c A M P bound. Repressor binding site in the operator is empty, but R N A polymerase does not bind. Transcription does not occur.

If the amount of glucose in the cell increases, the cAMP concentration falls, and without cAMP, CRP detaches from the *lac* operon. Because CRP is inactive, RNA polymerase binds less efficiently to the promoter, and transcription of the *lac* operon proceeds only at a low level, even when lactose is present **(Figure 18.5b)**. Thus, the *lac* operon is under dual control: negative control by the *lac* repressor and positive control by CRP. Whether or not transcription occurs is controlled by allolactose: Without allolactose, the *lac* repressor is active and the operon is off (transcription does not occur; see Figure 18.4a); with allolactose, the *lac* repressor is inactive and the operon is on (transcription occurs; see Figure 18.4b). The *rate* of transcription is controlled by whether CRP has cAMP bound to it: With bound cAMP, the rate is high; without it, the rate is low. It is as though the operon has both an on-off switch and a volume control.

In addition to regulating the *lac* operon, CRP helps regulate other operons that encode enzymes used in catabolic pathways. All told, it may affect the expression of more than 100 genes in *E. coli*. When glucose is plentiful and CRP is inactive, the synthesis of enzymes that catabolize compounds other than glucose generally slows down. The ability to catabolize other compounds, such as lactose, enables a cell deprived of glucose to survive. The compounds present in any given cell at a certain moment determine which operons are switched on—the result of simple interactions of activator and repressor proteins with the promoters of the genes in question.